pg. 129 Course Supplement

Grade: Check Plus

Assignment 2 - pg 2 Kelly Nobles 2/24/26

→ Question 1:

Price (x)	Probability f(x)	$x f(x)$	$(x-25.41)^2 * f(x)$
20	0.05	1.00	1.46
21	0.06	1.26	1.17
22	0.07	1.54	0.81
23	0.08	1.84	0.46
24	0.09	2.16	0.18
25	0.12	3.00	0.02
26	0.15	3.90	0.05
27	0.13	3.51	0.33
28	0.10	2.80	0.67
29	0.10	2.90	1.29
30	0.05	1.50	1.05
		$\boxed{\$25.41 = \mu}$	7.50

sum sum

✓ Q1

$\mu = \sum x \cdot f(x) = \boxed{\$25.41 \text{ mean stock price}}$

$\sigma = \sqrt{Var(s)} = \sum (x-\mu)^2 f(x)$ $\boxed{\sqrt{7.50} = 2.74 = \sigma}$

→ Question 2: Part A

$\mu = 25.41$ $\sigma = 2.74$

$25.41 - 2.74 = 22.67$ $25.41 + 2.74 = 28.15$

$P(22.67 \leq S \leq 28.15)$

✓ Q2a

Stock prices that fall between 22.67 and 28.15:

23	0.08
24	0.09
25	0.12
26	0.15
27	0.13
28	0.10
	0.67

sum

$\boxed{P \text{ w/in } 1 \text{ std. dev.} = 0.67}$

or

67% probability that stock price will be w/in 1 std. dev. of its mean 1yr. from now.